

Door Controller Wiring

Full interconnect for the ESP32-S3 fingerprint entry unit: signal wiring, power distribution, the lock drive stage, and the pin map your firmware should compile against.

SUPPLY 12 V DC · 5 A single adapter, fused 3 A	CONTROLLER ESP32-S3 N16R8 3.3 V logic, 16 MB / 8 MB PSRAM	UPLINK 4G LTE + Wi-Fi A7670C primary, Wi-Fi fallback
LOCK 12 V solenoid relay-switched, fail-secure		

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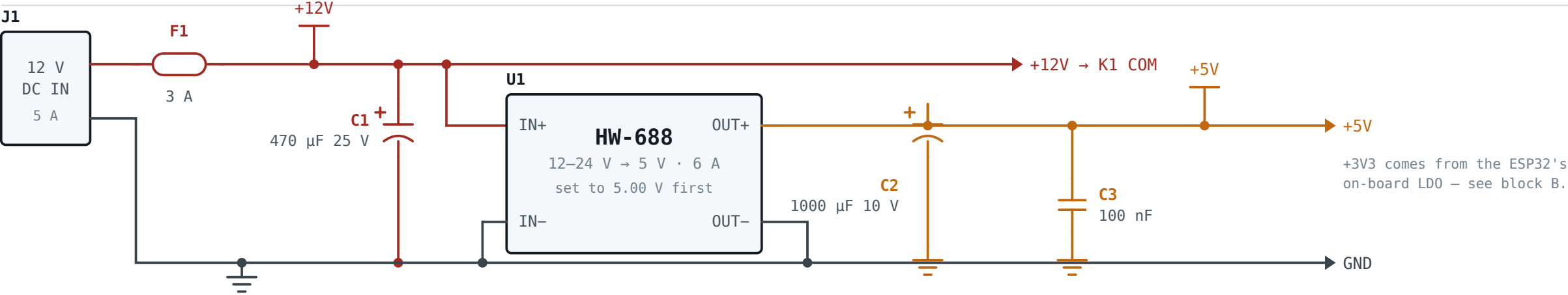
6. Firmware pin map

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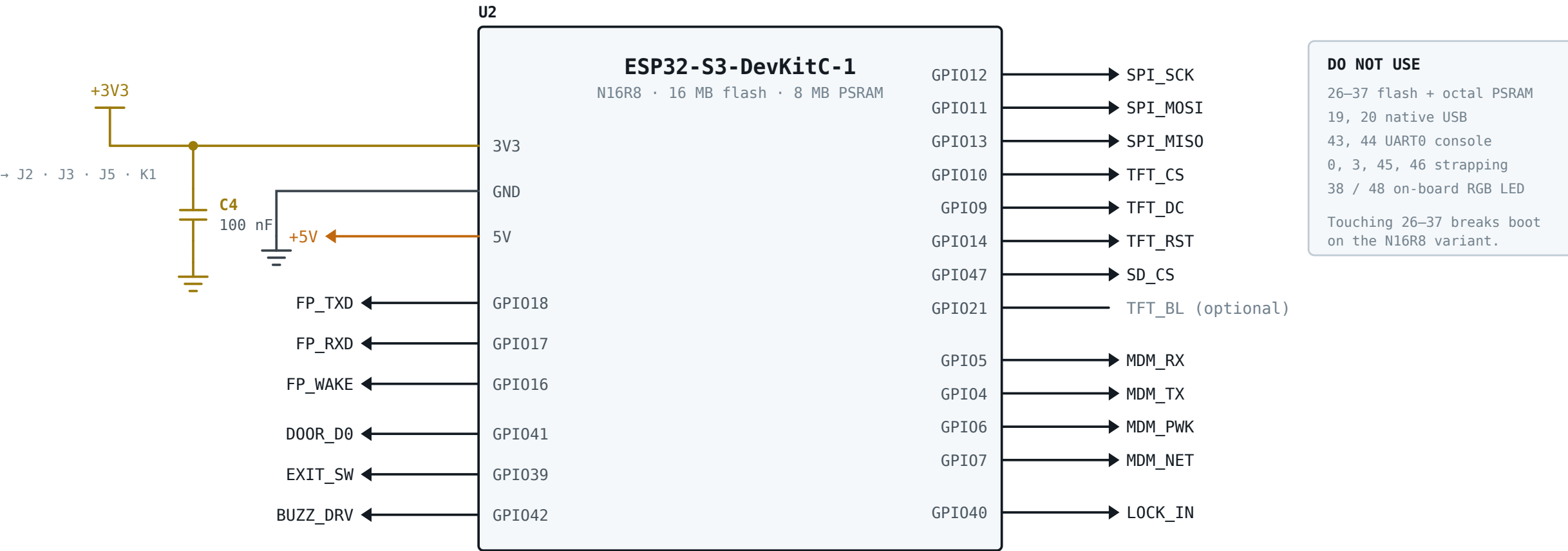
8. Bill of materials

Schematic and diagram pages are A3 landscape; tables and notes are A4 portrait. Reference designators on the sheets match the bill of materials in section 08.

A · POWER SUPPLY

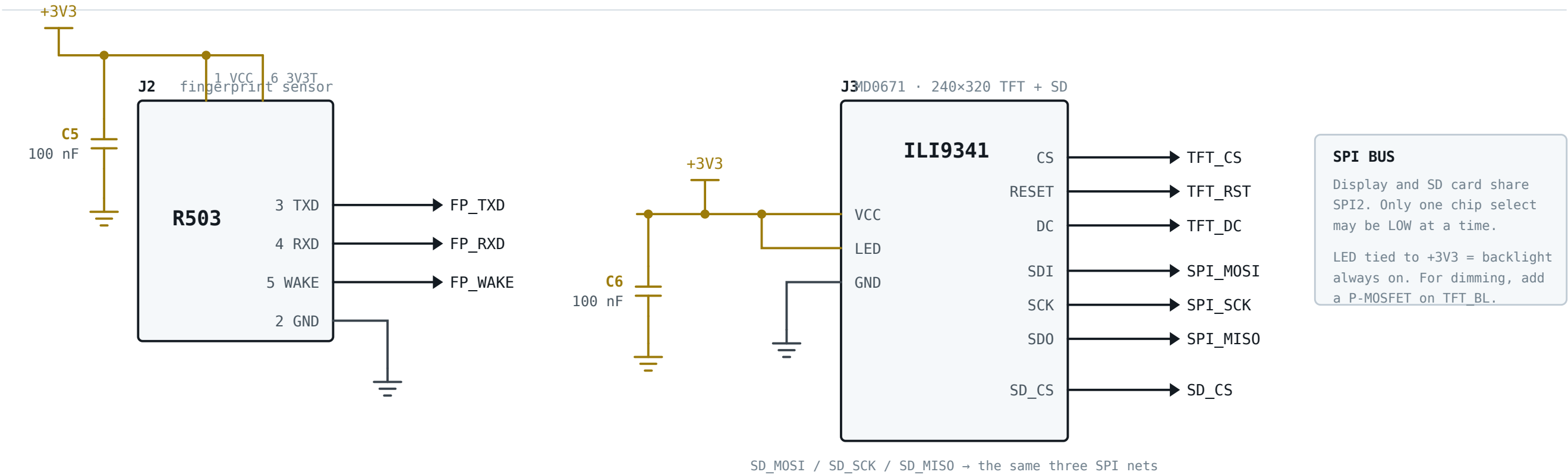


B · MCU

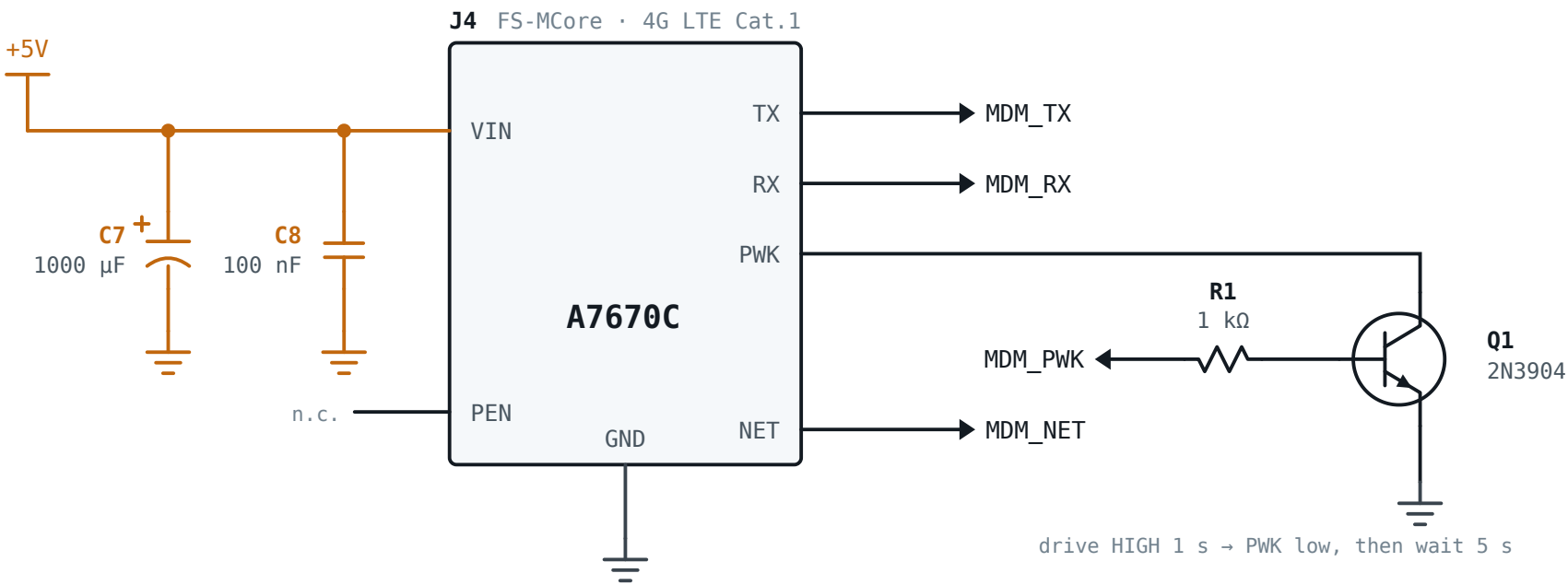


C · Fingerprint sensor & display · D · 4G modem

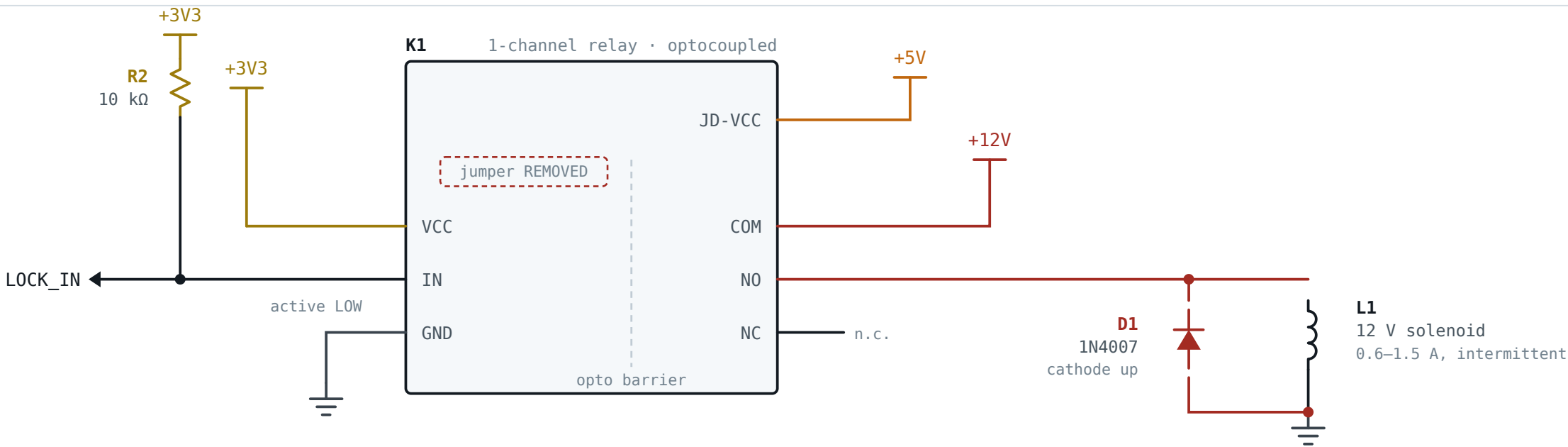
C · FINGERPRINT SENSOR & DISPLAY



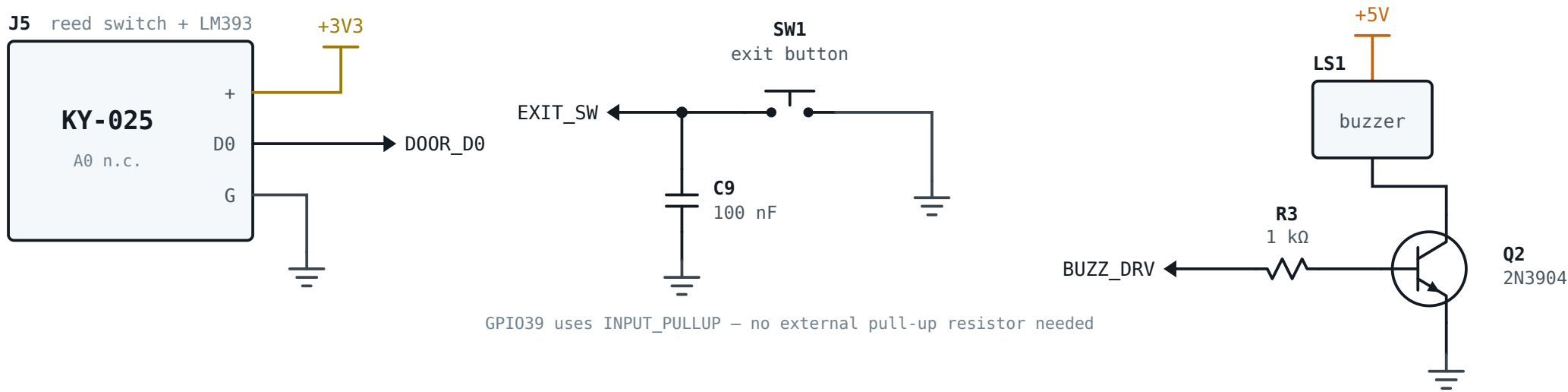
D · 4G MODEM



E · DOOR LOCK DRIVE



F · INPUTS & AUDIBLE FEEDBACK



NOVA FITNESS – Door Controller

Biometric access unit · schematic sheet 1 of 1

Rev A · ESP32-S3 N16R8 · 12 V single-supply · fail-secure lock
Verify connector orientation on your own modules before power-up.

SHEET NOTES

- 1 · Net labels with the same name are the same wire. Power flags (+12V, +5V, +3V3) and ground symbols are global.
- 2 · +3V3 is produced by U2's on-board LD0. Do not add 3.3 V loads beyond J2, J3, J5 and K1's opto input.
- 3 · J4's VIN and GND run as their own pair to U1's output terminal – not through U2 – because of 2.3 A LTE bursts.
- 4 · Remove K1's VCC+JD-VCC jumper. R2 keeps LOCK_IN HIGH through reset so the lock stays shut.

02 Signal overview

Every logic connection between the ESP32-S3 and the peripherals. Power rails are deliberately left out of this drawing — they get their own diagram in section 02, so neither picture ends up a tangle.

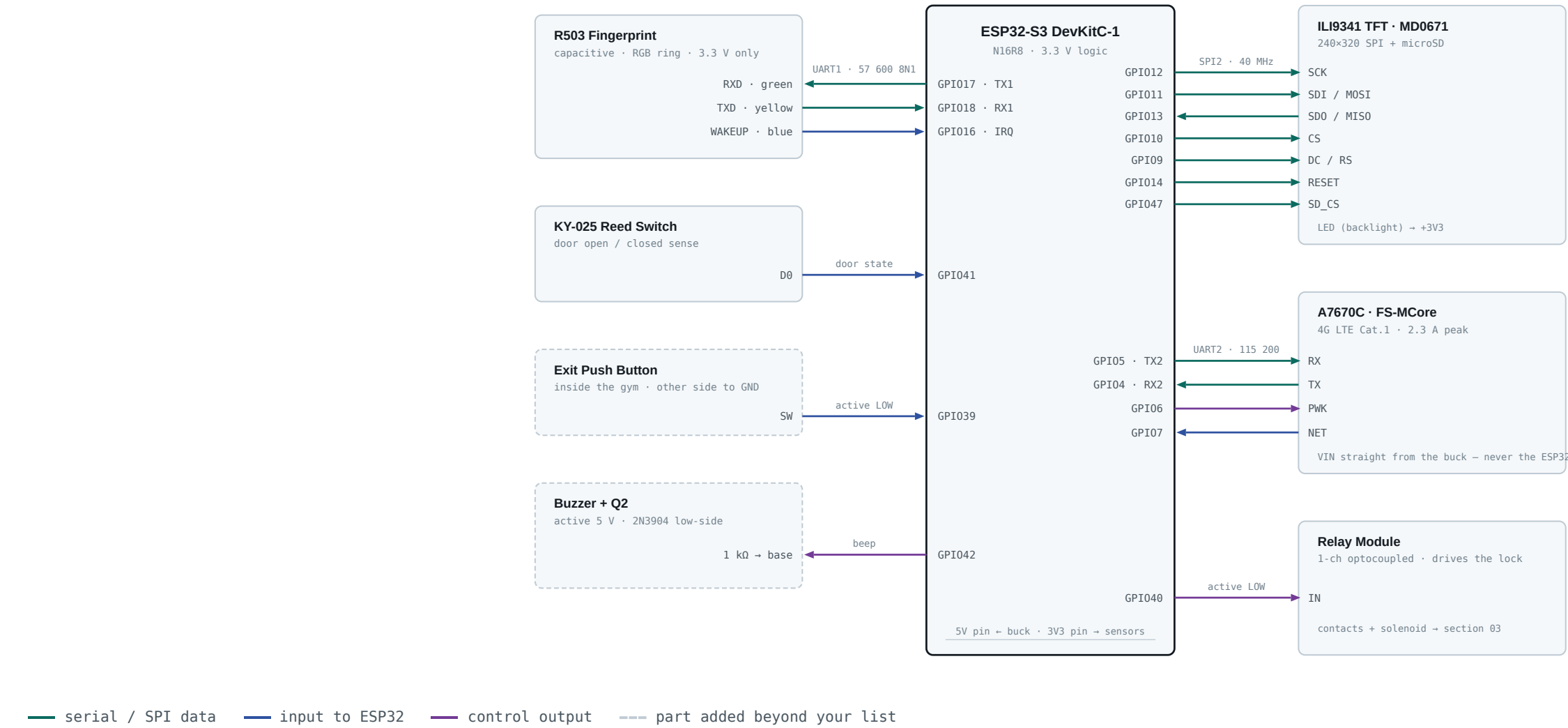


Fig. 1 — Signal interconnect. Arrows point in the direction data travels. Every GPIO shown is safe on the N16R8 variant: nothing here touches the pins the 16 MB flash and 8 MB octal PSRAM occupy (GPIO26–37), the native USB pins (19/20), the UART0 console (43/44) or the strapping pins (0, 3, 45, 46).

03 Power distribution

One 12 V adapter feeds two things: the lock, directly through the relay contacts, and the HW-688 buck. Everything digital hangs off the buck's 5 V output; the ESP32's own regulator makes the 3.3 V for the sensors.

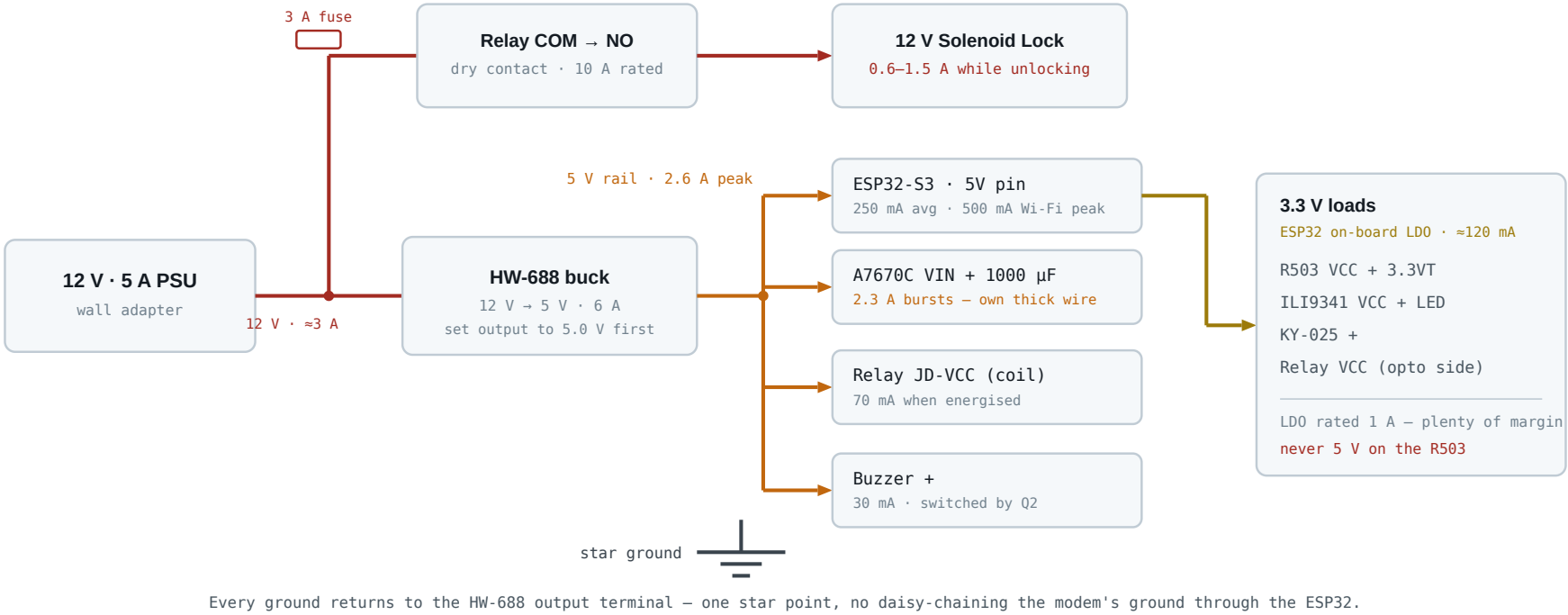


Fig. 2 — Power tree. The modem is the reason this diagram matters. A7670C transmit bursts pull well over 2 A for a few hundred microseconds at a time; if that current has to travel through the ESP32's 5 V pin and its ground pin, the resulting rail sag reboots the ESP32 mid-scan. Give the modem its own pair of wires from the buck's terminal block and a 1000 μF electrolytic right at its VIN.

RAIL	SOURCE	TYPICAL	PEAK	HEADROOM
12 V	Wall adapter	≈0.4 A idle	≈2.5 A unlocking	5 A adapter → comfortable. 3 A is the minimum.
5 V	HW-688 buck	≈0.65 A	≈2.6 A	6 A module → large margin, will barely warm.
3V3	ESP32 on-board LDO	≈120 mA	≈160 mA	1 A LDO → fine. Do not add more 3.3 V loads than these four.

04 Lock drive stage

The one part of the build where a wiring mistake costs you a board. Two details do all the work: the jumper comes off so the 3.3 V logic drives the optocoupler properly, and the pull-up keeps the door shut through boot and reset.

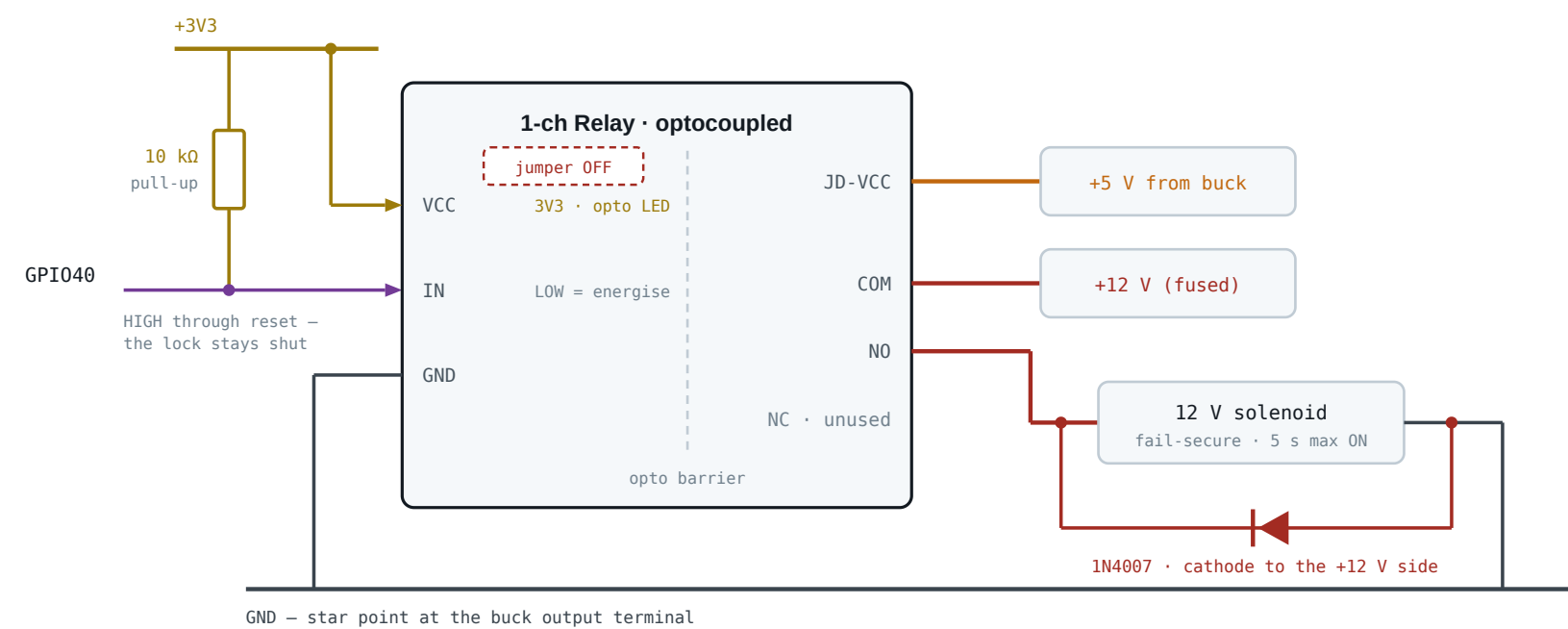


Fig. 3 — Relay and solenoid. Removing the VCC ↔ JD-VCC jumper splits the module into a 3.3 V input side, which the ESP32 can drive cleanly, and a 5 V coil side. Left jumpered at 5 V the optocoupler LED is marginal at 3.3 V and the relay chatters. The 10 kΩ pull-up matters just as much: without it the IN pin floats for the ~50 ms between reset and firmware start, and the door pops open every time the unit reboots.

Fail-secure vs fail-safe. A fail-secure solenoid stays locked with no power and only draws current while the relay is closed — that is the wiring drawn above and the right default for a gym. A fail-safe lock is the opposite: it is energised continuously to stay unlocked, drops open on power loss, and would keep 12 V flowing through your relay contacts 24/7. Check which one you bought before you power it up; if it is fail-safe, invert the relay logic in firmware and size the adapter for continuous draw.

05 Connection tables

Wire from these, not from the picture. Every row is one physical wire.

MODULE 02 R503 Fingerprint			MODULE 07 ILI9341 TFT · MD0671	
R503 PIN	WIRE	GOES TO	DISPLAY PIN	GOES TO
1 · VCC	red	3V3	VCC	3V3
2 · GND	black	GND	GND	GND
3 · TXD	yellow	GPI018	CS	GPI010
4 · RXD	green	GPI017	RESET	GPI014
5 · WAKEUP	blue	GPI016	DC / RS	GPI09
6 · 3.3VT	white	3V3	SDI / MOSI	GPI011
3.3 V only — 5 V on pin 1 destroys the sensor. Default baud 57 600 8N1. WAKEUP goes active when a finger touches the ring, so the firmware can sit in a light sleep and wake on interrupt instead of polling. Wire colours vary between batches: confirm against pin 1 on your connector, not the colour.			SCK	GPI012
			LED	3V3 (always on)
			SDO / MISO	GPI013
			SD_CS	GPI047
			SD_MOSI / SCK / MISO	tie to the same GPIO11 / 12 / 13
			Run the whole board at 3.3 V — these modules have no level shifters, and 5 V on the logic pins is what kills them. For a dimmable backlight, drive LED through a PNP or P-MOSFET from 3V3 with the gate on GPI021; do not drive it straight from a GPIO, it draws far more than a pin can source.	

MODULE 03

A7670C · FS-MCore

BOARD PIN	GOES TO
VIN	5V from the buck
GND	GND at the buck terminal
TX	GPI04
RX	GPI05
PWK	GPI06 via NPN, or a jumper
NET	GPI07 (optional)
PEN	leave open

Power-on sequence: pull PWK low for ~1 s, release, then wait ~5 s before the first AT command. Cleanest way is an NPN (2N3904, 1 kΩ on the base) pulling PWK to GND, so the ESP32 never has to source into the modem's rail. This board has on-board level translation, so TX/RX go straight to the ESP32 — but check your silkscreen for a VDD/VREF pin; if there is one, tie it to 3V3.

MODULES 05 + 04

Relay + Solenoid Lock

RELAY PIN	GOES TO
VCC	3V3
GND	GND
IN	GPI040 + 10 kΩ to 3V3
JD-VCC	5V
VCC ↔ JD-VCC jumper	removed
COM	+12 V after the fuse
NO	solenoid +
NC	unused

Solenoid – goes to 12 V ground, with a 1N4007 across the coil, cathode to the + side. Keep the unlock pulse to 3–5 s in firmware — most 12 V solenoids are rated for intermittent duty and will cook if held.

MODULE 06

KY-025 Reed Switch

KY-025 PIN	GOES TO
+ / VCC	3V3
G / GND	GND
D0	GPI041
A0	unused

Power it from 3.3 V, not 5 V — the LM393 comparator works down to 3 V, and this keeps D0 at a level the ESP32 can read safely. At 5 V you would need a divider. Set the trim pot with the magnet in the closed-door position. Sensor on the frame, magnet on the door leaf.

ADDED PARTS

Exit Button + Buzzer

PART	WIRING
Exit button	GPI039 → switch → GND , INPUT_PULLUP , 100 nF across the switch
Buzzer +	5V
Buzzer -	collector of Q2 (2N3904)
Q2 base	GPI042 via 1 kΩ
Q2 emitter	GND

An active 5 V buzzer only needs a level on the pin — no tone generation. If you buy a passive one, drive Q2 from an LEDC PWM channel instead. Run the exit button on its own twisted pair if the cable run is over a metre.

06 Firmware pin map

Paste this into a `pins.h` and you are done arguing with the pinout.

GPI O	SIGNAL	DIRECTION	PERIPHERAL	NOTES
4	MODEM_RX	in	UART2	from A7670C TX
5	MODEM_TX	out	UART2	to A7670C RX
6	MODEM_PWRKEY	out	GPIO	via NPN, 1 s low pulse
7	MODEM_NET	in	GPIO	network status, optional
9	TFT_DC	out	GPIO	data / command select
10	TFT_CS	out	SPI2	display chip select
11	SPI_MOSI	out	SPI2	shared: display + SD
12	SPI_SCK	out	SPI2	shared: display + SD
13	SPI_MISO	in	SPI2	SD card reads
14	TFT_RST	out	GPIO	display reset
16	FP_TOUCH	in	GPIO / IRQ	R503 WAKEUP
17	FP_TX	out	UART1	to R503 RXD
18	FP_RX	in	UART1	from R503 TXD
21	TFT_BL	out	LEDC	reserved — only if you fit the backlight driver
39	EXIT_BTN	in	GPIO	<code>INPUT_PULLUP</code> , active low
40	LOCK_RELAY	out	GPIO	active low, external 10 k pull-up
41	DOOR_SENSE	in	GPIO	KY-025 D0
42	BUZZER	out	GPIO / LEDC	via 2N3904
47	SD_CS	out	SPI2	microSD chip select

Pins deliberately left alone on the N16R8: 26–37 are wired to the 16 MB flash and 8 MB octal PSRAM and will brick the boot if you touch them. 19/20 are the native USB D–/D+. 43/44 are the UART0 console you will be watching logs on. 0, 3, 45, 46 are strapping pins that decide boot mode. 38 and 48 carry the board's RGB LED depending on your revision.

07 Before you power it up

Ordered roughly by how expensive the mistake is.

KILLS HARDWARE Set the buck output before connecting anything

Wire the HW-688 to 12 V with nothing on its output, meter the output terminals, adjust to 5.0 V, then power down and connect the loads. Some of these modules ship set near their input voltage — connecting the ESP32 first is a very short experiment.

KILLS HARDWARE USB and 12 V at the same time

On the DevKitC-1 the USB VBUS and the 5V header pin sit on the same net with no ORing diode. While flashing over USB, disconnect the 12 V supply — or fit a Schottky (SS34) in the buck's 5 V line so the two sources cannot fight.

RESETS MID-SCAN Modem inrush

The 1000 μ F at the A7670C VIN plus a 100 nF ceramic beside it is not optional in practice. Without it, an LTE transmit burst drags the shared 5 V rail down far enough to brown out the ESP32, and it shows up as random reboots that look like a firmware bug for about two days.

OPENS THE DOOR Relay state at boot

Fit the 10 k Ω pull-up on GPIO40 and set the pin HIGH with `pinMode/digitalWrite` as the very first lines in `setup()`, before Wi-Fi, display or anything else. Confirm on the bench that a reset does not click the relay.

CONTACT WEAR Flyback diode across the solenoid

Switching an inductive 12 V load with no diode arcs the relay contacts and radiates enough noise to corrupt the SPI display. A 1N4007 costs almost nothing and is the difference between a relay that lasts years and one that welds shut. Mount it at the lock, not at the relay.

WIRING Separate the lock cable from the signal cable

Run the 12 V lock pair in its own sleeve, away from the R503 and display ribbons. Keep the fingerprint cable under 30 cm if you can; if the sensor has to sit further out, drop the UART to 9600 baud rather than fighting the noise.

ANTENNA Give the A7670C room

Mount the LTE antenna outside the metal enclosure and keep it away from the buck's inductor and the solenoid wiring. Insert the SIM before power-up — the module will not detect a hot-swapped card.

BRING-UP ORDER Test one module at a time

Buck → ESP32 alone → display → R503 → relay with no lock attached (listen for the click) → lock → modem last. Each step is a few minutes; debugging all seven at once is not.

08 Bill of materials

Every reference designator on the schematic sheet, in one list. The modules you already own are at the top; everything below the rule is what has to be added to make the sheet buildable.

REF	PART	VALUE / TYPE	QTY	PURPOSE
U2	ESP32-S3 DevKitC-1	N16R8	1	Controller
U1	HW-688 buck module	12–24 V → 5 V, 6 A	1	5 V rail
J2	R503 fingerprint sensor	capacitive, RGB ring	1	Biometric read
J3	ILI9341 TFT + SD	MD0671, 2.4" 240×320	1	User display
J4	A7670C modem	FS-MCore LTE Cat.1	1	4G uplink
J5	KY-025 reed module	reed + LM393	1	Door open sense
K1	Relay module	1-ch, 5 V coil, optocoupled	1	Lock switching
L1	Solenoid door lock	12 V, fail-secure	1	The lock
ADDED TO MAKE THE CIRCUIT WORK				
J1	DC barrel jack + 12 V adapter	12 V, 5 A	1	Supply
F1	Fuse + holder	3 A slow-blow	1	A shorted lock cable otherwise takes the adapter with it
C1	Electrolytic	470 μF / 25 V	1	12 V input bulk
C2	Electrolytic	1000 μF / 10 V	1	5 V rail bulk at the buck output

REF	PART	VALUE / TYPE	QTY	PURPOSE
C7	Electrolytic	1000 μ F / 10 V	1	At J4 VIN — absorbs the 2.3 A LTE bursts
C3–C6, C8, C9	Ceramic	100 nF / 50 V	6	Local decoupling + switch debounce
R1, R3	Resistor	1 k Ω $\frac{1}{4}$ W	2	Transistor base current limit
R2	Resistor	10 k Ω $\frac{1}{4}$ W	1	Holds LOCK_IN high through reset — the lock stays shut
Q1, Q2	NPN transistor	2N3904 (or BC547)	2	Modem PWRKEY pull-down; buzzer low-side drive
D1	Rectifier diode	1N4007	1	Solenoid flyback clamp — protects the relay contacts
SW1	Momentary push button	NO, panel mount	1	Exit from inside without a fingerprint
LS1	Active buzzer	5 V, ~30 mA	1	Accept / deny feedback
—	Screw terminal blocks	2-way, 5 mm	4	12 V in, buck output star point, lock, sensor tails

Substitutions that are fine: any 470–1000 μ F electrolytic rated 16 V or more for C1; BC547 or S8050 for the 2N3904s; 1N4001–1N4007 for D1; 4.7 k Ω –10 k Ω for R2. **Not fine:** dropping C7, dropping D1, or leaving R2 out — those three are what stop the unit rebooting mid-scan, welding the relay contacts, and opening the door on every reset.

Rev A · drawn for the NOVA FITNESS access controller, stage 13. Verify the R503 connector orientation and the A7670C silkscreen against your own boards before applying power — module revisions differ. Datasheet references: [GROW R503 manual](#), [SIMA7670C FS-MCore](#).